

VISWANATH VASA

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EDUCATION

University of Maryland – College Park

B.S. Computer Science (Cybersecurity) & Applied Mathematics | Honors College

College Park, MD

Expected May 2027

- Banneker/Key Full Merit Scholar

- Relevant Coursework: Data Structures, Computer Networks, Operating Systems, Cybersecurity, Probability & Statistics

CompTIA Network+ Certification

Network Administration, Security Fundamentals, Infrastructure, IoT Systems

Mount Airy, MD

TECHNICAL SKILLS

Programming & Development: Python, JavaScript, C, SQL, React, Node.js

Systems & Networking: Network Administration, TCP/IP, DNS, IoT Systems, Device Provisioning

Cloud & DevOps: Docker, CI/CD, Google Cloud Platform, AWS

Security: Cybersecurity Fundamentals, Authentication, Role-Based Access Control, Secure APIs

Data & Analytics: Machine Learning, Statistical Analysis, Data Pipelines

EXPERIENCE

Researcher & Lead Full-Stack Developer

Empowering Secure Elections Laboratory

May 2025 – Present

College Park, MD

- Engineered a secure full-stack **Flutter** mobile application with a **Firebase backend**, implementing **authentication, role-based access control**, and encrypted client-server communication for **2,000+ users**.
- Designed an **offline-first** SQLite data pipeline with **asynchronous cloud synchronization**, ensuring reliable **data integrity**, auditability, and fault tolerance for **1,000+ user accounts**.
- Developed a **modular** content delivery system supporting dynamic text, images, and video, enabling non-technical teams to deploy **50+ interactive training modules**.

Frontend Developer (Contract)

Sunrise Scientific

Dec. 2025 – Present

Towson, MD

- Designed and implemented **information architecture** and **navigation flows** across **15+ pages**, improving usability and content discoverability for enterprise customers.
- Iterated on **UI layouts** and **interaction patterns** based on stakeholder feedback, enhancing **customer experience** and supporting engagement across **100+ active users**.

Machine Learning Engineer

Haridra Solutions, Inc.

Jun. 2023 – Aug. 2024

Mount Airy, MD

- Engineered **machine learning models in Python** to support language learning applications, contributing to a **30% increase** in student applications to top-tier colleges.
- Optimized **model inference performance** by over **20%**, reducing processing latency from **45s to 35s per query** and improving system efficiency.

PROJECTS

SenseGuard, Inc. | React, Node.js, PostgreSQL, Docker, Google Cloud Platform

Sep. 2025 – Present

- Designed and deployed a **full-stack security monitoring platform** processing **1,000+ telemetry events/day** across **distributed IoT systems**.
- Containerized backend services using **Docker** and deployed to **Google Cloud Platform**, supporting **CI/CD workflows** and maintaining **99.9% service availability**.
- Configured and provisioned **200+ IoT sensors and gateways** across **50+ sites**, validating end-to-end data pipelines and standardizing **device onboarding procedures**.

Autonomous Robotics Optimization | C, ARM Assembly, FTC SDK

Sep. 2021 – Aug. 2024

- Optimized **autonomous navigation algorithms** using **C and ARM-specific optimizations**, achieving **25%+ performance improvement** in real-time systems.
- Implemented a custom **network communication protocol** using the **FTC SDK**, improving data transfer efficiency by **75%**.